Violating the Assumptions of the Classical Linear Regression Model

These notes accompany Lecture 6. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Lectures 2 and 3 built up a set of assumptions under which OLS is unbiased, efficient, and supports valid inference. This lecture asks what happens when each one fails – how to detect the failure, what it costs, and what can be done about it.

Learning outcomes – after this lecture you should be able to:

1. Distinguish exogenous from endogenous sample selection, and explain why only the second causes bias.
2. Explain what multicollinearity does and does not do to OLS, and interpret a variance inflation factor.
3. State the consequences of heteroskedasticity – separating what is affected from what is not.
4. Carry out and interpret the Breusch-Pagan-Godfrey and White tests.
5. Explain what heteroskedasticity-robust standard errors fix, and what they do not.
6. Test residuals for normality and explain why large samples make the assumption less critical.
7. Identify influential observations and handle them with dummy variables.

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1. Nonrandom samples (MLR.2)
2. Multicollinearity (MLR.3)
3. Heteroskedasticity (MLR.5)
4. Normality of the residuals (MLR.6)
5. Summary of violations

The assumptions, for reference

Assumption	Statement
MLR.1 Linear in parameters	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k\, x_k + u$
MLR.2 Random sampling	A random sample of n observations from the population
MLR.3 No perfect collinearity	No regressor is constant and there are no exact linear relationships among them
MLR.4 Zero conditional mean	$E(u \mid x_1, \ldots, x_k) = 0$
MLR.5 Homoskedasticity	$Var(u \mid x_1, \ldots, x_k) = \sigma^2$
MLR.6 Normality	$u \sim Normal(0, \sigma^2)$

This lecture covers MLR.2, MLR.3, MLR.5 and MLR.6. MLR.4 – the zero conditional mean, whose failure is called endogeneity – is the most important of them all and is the subject of Lecture 7.

1. Nonrandom samples (MLR.2)

If the random sampling assumption fails, we must ask what it costs. The answer turns entirely on one question: was the sample selected on the basis of the explanatory variables, or on the basis of the dependent variable?

1.1 Exogenous sample selection – usually harmless

Suppose we estimate a saving function:

$$saving = \beta_0 + \beta_1\, income + \beta_2\, age + \beta_3\, size + u$$

and our data come from a survey of people over 35 years of age. This is a nonrandom sample of all adults – selected on age, which is one of the regressors.

This is not ideal, but we can still obtain unbiased and consistent estimators of the population parameters. The reason is that the regression function is the same for any subset of the population defined by income, age or size: conditioning on a regressor does not disturb E(u | x) = 0.

> **The intuition, and its limit**
>
> You have estimated the slope of a line using only the right-hand portion of the data. If the line really is straight, the slope you recover is correct everywhere.
>
> If the relationship actually changes shape across the range – if the true return to income differs for the under-35s – then you will be wrong on the left-hand side. Exogenous selection is safe only to the extent that the model is correctly specified over the whole population.
>
> Provided there is enough variation in the regressors within the subpopulation, the only real cost is a smaller sample size.

1.2 Endogenous sample selection – always biased

The situation is entirely different when selection depends on y. Suppose we want the relationship between individual wealth and other factors in the population of all adults:

$$wealth = \beta_0 + \beta_1\, educ + \beta_2\, exper + \beta_3\, age + u$$

but only people with wealth below $250,000 are included in the sample. This is a nonrandom sample based on the value of the dependent variable, and it produces biased and inconsistent estimators – because the population regression function is not the same as the expected value conditional on wealth being under $250,000.

> **Why selecting on y is selecting on u**
>
> Since y = xβ + u, choosing observations by their value of y means choosing simultaneously on x and on u.
>
> Suppose y = βx + u and you keep only observations with low y. Then observations with high x can only survive if they have an unusually low u, while observations with low x survive regardless.
>
> The surviving sample therefore has Cov(x, u) ≠ 0 and E(u | x, selected) ≠ 0 – MLR.4 fails as a direct consequence of how the sample was built.
>
> This is why no amount of data fixes it: the bias is in the selection rule, not in the sample size.

2. Multicollinearity (MLR.3)

MLR.3 forbids exact linear relationships among the regressors. Software will normally refuse to estimate a perfectly collinear model, so the more common and realistic case is regressors that are strongly but imperfectly correlated.

High but not perfect correlation between two or more independent variables is called **multicollinearity**. A standard example is a consumption function including both income and wealth, which are likely to move together closely.

2.1 What it does

Recall the sampling variance from Lecture 2:

$$Var(\hat{\beta}_j) = \sigma^2 / [\, SST_j(1 - R_j^2) \,]$$

As x_j becomes more strongly related to the other regressors, R_j^2 rises towards one and the variance grows without bound. Adding an irrelevant variable to an equation generally increases the variances of the remaining estimators for exactly this reason.

> **The crucial point about multicollinearity**
>
> Multicollinearity violates none of the MLR assumptions. OLS remains unbiased and remains BLUE.
>
> The "problem" of multicollinearity is therefore not really well defined – it is a matter of degree, not a violation. What is clear is only this: other things equal, for estimating β_j it is better to have less correlation between x_j and the other regressors.

The practical symptoms follow from the inflated variances:

- Estimated coefficients become statistically insignificant, even when the group of variables is jointly significant.
- Confidence intervals become wide – the estimates are imprecise.
- Results become sensitive to small changes in the data or the specification.

Notice what is missing from that list: bias. If the coefficients are insignificant but correctly centred, more data would solve the problem. That is the sense in which multicollinearity is a data deficiency rather than a modelling error.

2.2 Measuring it: the variance inflation factor

The **variance inflation factor** is the factor by which the variance of $\hat{\beta}_j$ is raised because x_j is not uncorrelated with the other regressors:

$$VIF_j = 1 / (1 - R_j^2)$$

In EViews: View → Coefficient Diagnostics → Variance Inflation Factors, reading the centred VIFs.

A cutoff above which multicollinearity is declared a "problem" is arbitrary and not especially helpful. The value 10 is sometimes used – corresponding to $R_j^2 = 0.9$ – but it is easy to misuse such a statistic, because we cannot say how much correlation among regressors is too much.

2.3 What to do about it

Option	Comment
Do nothing	Often there is no choice – the data simply lack independent variation. This is a legitimate answer.
Drop a variable	Reduces the variance, but risks omitted variable bias. You are trading a known imprecision for a possible bias.
Transform the variables	First differences, ratios, or other reparameterisations can break the collinearity.
Principal components (PCA)	Replaces many correlated regressors with a few uncorrelated linear combinations retaining most of the variation – at the cost of interpretability.

3. Heteroskedasticity (MLR.5)

3.1 What it means

Both words split into two Greek parts: homo (same, equal) or hetero (different, unequal), plus skedastic (spread, scatter). So homoskedasticity means equal spread and heteroskedasticity unequal spread. Since the measure of spread in econometrics is the variance, heteroskedasticity means unequal variances.

MLR.5 requires $\text{Var}(u \mid x_1, \ldots, x_k) = \sigma^2$. Under heteroskedasticity the variance of the error depends on which observation we are discussing.

> **The standard example: income and consumption**
>
> People on low incomes have little flexibility in spending. A large share goes on food and transport, so consumption patterns differ little between them and the spread around the regression line is small.
>
> Richer people have far more choice. Some consume heavily, others save or invest, so actual consumption can be far from average consumption at that income.
>
> The spread therefore grows with income – a textbook case of $\text{Var}(u_i) = \sigma_i^2$ rather than a common σ^2.

3.2 Consequences for OLS

It is worth setting out precisely what does and does not break.

Still fine	No longer valid
Unbiasedness of the $\hat{\beta}_j$	The estimated variances $\text{Var}(\hat{\beta}_j)$ are biased
Consistency of the $\hat{\beta}_j$	OLS standard errors are invalid
Interpretation of R^2 and adjusted R^2	t-statistics no longer have t-distributions
–	F-statistics are no longer F-distributed
–	OLS is no longer BLUE

Homoskedasticity played no part in proving unbiasedness or consistency – those came from MLR.1 through MLR.4 alone. What MLR.5 delivers is the validity of the variance formula, and everything built on it.

Crucially, the problem is not resolved by a larger sample. A bigger n gives a more precise estimate of the wrong standard error.

3.3 The Breusch-Pagan-Godfrey LM test

Breusch and Pagan (1979) developed a Lagrange multiplier test. Starting from $y = \beta_o + \beta_1 x_1 + \ldots + \beta_k$ x_k + u:

1. Estimate the model by OLS and obtain the residuals $\hat{u}_i$.
2. Run the auxiliary regression of the squared residuals on all the original explanatory variables: $\hat{u}_i^2 = \delta_o + \delta_1 x_1 + \ldots + \delta_k$ x_k + v.
3. Formulate the hypotheses. H_o: $\delta_1 = \delta_2 = \ldots = \delta_k = 0$ (the variance does not depend on the regressors – homoskedasticity) against H_1: at least one $\delta_j \neq 0$.
4. Compute $LM = n \cdot R^2$, where n and R^2 come from the auxiliary regression. Under H_o this follows a χ^2 distribution with k degrees of freedom.
5. If the p-value is below the chosen significance level, reject homoskedasticity.

In EViews: View → Residual Diagnostics → Heteroskedasticity Tests → Breusch-Pagan-Godfrey.

For the wage model the result is $p = 0.106 > 0.05$, so we fail to reject H_o: there is no strong evidence of non-constant variance.

3.4 The White test

White (1980) proposed a more general test, adding the squares and all cross products of the regressors to the auxiliary equation. With k = 3 regressors the auxiliary regression contains x_1, x_2, x_3, their three squares and their three cross products – nine terms plus the intercept.

The test statistic is again the LM statistic for the hypothesis that all the auxiliary slope coefficients are zero.

Because it is more general, the White test is usually recommended over the alternatives. The practical drawback is that the cross-product terms make the number of regressors grow quickly, which costs degrees of freedom.

For the wage model, $p = 0.129 > 0.05$ – again we fail to reject homoskedasticity.

3.5 Heteroskedasticity-robust inference

White (1980) also proposed a way of obtaining consistent estimators of the variances and covariances of the OLS estimators without knowing the form of the heteroskedasticity – or indeed whether any is present.

These are variously called White, Huber or Eicker standard errors; we will call them **heteroskedasticity-robust standard errors**, or simply robust standard errors.

In EViews: in the Equation Estimation window click Options, then under Covariance method choose Huber-White.

What robust standard errors do and do not change

They change only the standard errors, t-statistics, and F-statistics. The coefficient estimates are exactly the same – this is not an alternative estimator, it is an alternative variance formula for the same estimator.

Robust standard errors can be either larger or smaller than the usual ones, though in practice they are more often larger.

EViews reports the robust joint test as the Wald F-statistic at the foot of the output.

In the wage application, most variables significant under the usual t-statistics remain significant under the robust ones; the coefficient on singfem becomes insignificant at 5%.

Wage model without SE correction

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/06/26 Time: 13:51
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.321378	0.100009	3.213492	0.0014
MARMALE	0.212676	0.055357	3.841881	0.0001
MARFEM	-0.198268	0.057835	-3.428132	0.0007
SINGFEM	-0.110350	0.055742	-1.979658	0.0483
EDUC	0.078910	0.006694	11.78733	0.0000
EXPER	0.026801	0.005243	5.111835	0.0000
EXPER^2	-0.000535	0.000110	-4.847105	0.0000
TENURE	0.029088	0.006762	4.301614	0.0000
TENURE^2	-0.000533	0.000231	-2.305553	0.0215

R-squared	0.460877	Mean dependent var	1.623268
Adjusted R-squared	0.452535	S.D. dependent var	0.531538
S.E. of regression	0.393290	Akaike info criterion	0.988423
Sum squared resid	79.96799	Schwarz criterion	1.061403
Log likelihood	-250.9552	Hannan-Quinn criter.	1.016998
F-statistic	55.24559	Durbin-Watson stat	1.784785
Prob(F-statistic)	0.000000		

Wage model with robust SE

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/06/26 Time: 13:52
Sample: 1 526
Included observations: 526
Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.321378	0.109469	2.935791	0.0035
MARMALE	0.212676	0.057142	3.721886	0.0002
MARFEM	-0.198268	0.058770	-3.373619	0.0008
SINGFEM	-0.110350	0.057116	-1.932028	0.0539
EDUC	0.078910	0.007415	10.64246	0.0000
EXPER	0.026801	0.005139	5.215010	0.0000
EXPER^2	-0.000535	0.000106	-5.033361	0.0000
TENURE	0.029088	0.006941	4.190731	0.0000
TENURE^2	-0.000533	0.000244	-2.187835	0.0291

R-squared	0.460877	Mean dependent var	1.623268
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Log likelihood	-250.9552	Hannan-Quinn criter.	1.016998
F-statistic	55.24559	Durbin-Watson stat	1.784785
Prob(F-statistic)	0.000000	Wald F-statistic	51.69553
Prob(Wald F-statistic)	0.000000		

Then why ever use the usual standard errors?

If homoskedasticity holds and the errors are normal, the usual t-statistics have exact t-distributions at any sample size.

Robust standard errors are justified only asymptotically. In small samples the robust t-statistics can have distributions that are not close to the t-distribution.

So in large samples, default to robust. In small samples, the choice is genuinely a trade-off.

4. Normality of the residuals (MLR.6)

4.1 Why it matters

If u is not normally distributed then the $\hat{\beta}_j$ are not normally distributed either, so t-statistics do not have t-distributions and F-statistics do not have F-distributions. This is potentially serious, since all our inference draws critical values and p-values from precisely those distributions.

Examining the residuals is also one of the most useful ways of detecting misspecification more generally.

4.2 The Jarque-Bera test

The Jarque-Bera statistic tests the null that the residuals are normally distributed, by comparing their skewness and kurtosis with the values implied by the normal distribution.

In EViews: View → Residual Diagnostics → Histogram – Normality Test. If the p-value is below 0.05, reject normality at the 5% level.

For the wage model the histogram is visibly non-normal and p = 0.00, so we reject the null. That sounds alarming – but note the sample size, over 500 observations, which brings us to the next point.

4.3 Asymptotic normality – why large samples rescue us

By the central limit theorem the OLS estimators satisfy asymptotic normality: they are approximately normally distributed in large enough samples. Testing and confidence intervals then proceed exactly as under the CLM assumptions.

This is valuable because MLR.6 has been dropped entirely. The only restriction now placed on the error distribution is that it have finite variance – that the errors are not wildly explosive – which we always assume. Zero conditional mean (MLR.4) and homoskedasticity (MLR.5) are still required.

> **How large is large enough?**
>
> There are no general prescriptions. Some econometricians consider n = 30 satisfactory, but that cannot suffice for every possible error distribution.
>
> In practice, more than 100 observations is a reasonable working benchmark for cross-sections.
>
> This is why rejecting Jarque-Bera in a sample of 526 is much less worrying than rejecting it in a sample of 32 – as in the example that follows.

4.4 Outliers and influential observations

An observation is **influential** if dropping it changes the key OLS estimates by a practically large amount. OLS is particularly exposed to such observations because it minimises the sum of *squared* residuals, so large residuals receive disproportionate weight.

Outliers commonly arise when sampling from a small population in which one or several members differ markedly from the rest. They can also produce non-normal residuals – which is why this topic sits under MLR.6.

R&D intensity and firm size

Relate R&D expenditure as a percentage of sales (rdintens) to sales (in millions) and profits as a percentage of sales (profmarg), using 32 chemical companies:

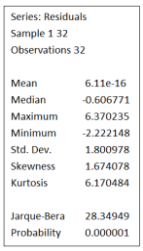
$$rdintens = \beta_0 + \beta_1\, sales + \beta_2\, profmarg + u$$

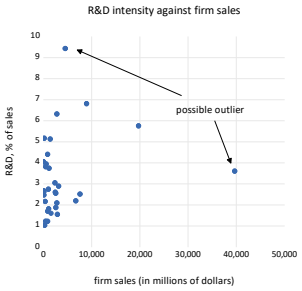
In the baseline regression neither sales nor profmarg is significant even at 10%, and the residuals are not normal. Of the 32 firms, 31 have annual sales below $20 billion; one has almost $40 billion – over twice the size of every other firm.

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:31
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.625261	0.585533	4.483542	0.0001
SALES	5.34E-05	4.41E-05	1.211091	0.2356
PROFMARG	0.044617	0.046181	0.966134	0.3420

R-squared	0.076118	Mean dependent var	3.266210
Adjusted R-squared	0.012403	S.D. dependent var	1.873701
S.E. of regression	1.862046	Akaike info criterion	4.170289
Sum squared resid	100.5492	Schwarz criterion	4.307701
Log likelihood	-63.72462	Hannan-Quinn criter.	4.215837
F-statistic	1.194653	Durbin-Watson stat	1.694228
Prob(F-statistic)	0.317275		





The dummy variable trick

Rather than deleting the observation, create a dummy equal to one for that observation and zero for all others, and include it alongside the other regressors. The dummy absorbs that observation entirely, so the remaining coefficients are estimated as if it had been dropped – but the coefficient on the dummy tells you how far the observation sits from the line fitted to the rest, and its t-statistic tells you whether that distance is large relative to its standard deviation.

The sequence of results is worth reading as a whole:

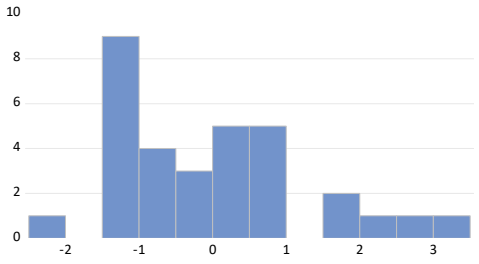
Specification	sales	profmarg	Dummy coefficients	R^2
Baseline (all 32 firms)	0.0000534 (p = 0.24)	0.045 (p = 0.34)	–	0.076
+ dummy for firm 10 (largest)	0.000186 (p = 0.04)	0.048 (p = 0.29)	D_{10} = −6.57 (p = 0.08)	0.174
+ dummy for firm 1 (highest rdintens)	0.0000506 (p = 0.15)	0.068 (p = 0.07)	D_1 = 6.72 (p = 0.00)	0.469
+ both dummies	0.000160 (p = 0.02)	0.070 (p = 0.05)	D_1 = 6.47, D_{10} = −5.41	0.535

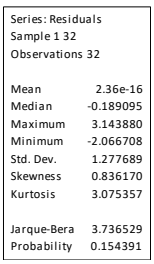
Reading across: excluding the largest firm more than triples the coefficient on sales and pushes its t-statistic above two, so within the sample of smaller firms there is a statistically significant positive relationship between R&D intensity and firm size. Excluding firm 1 instead barely moves the sales coefficient but makes profmarg larger and marginally significant. With both dummies, both economic variables are significant, and normality of the residuals can no longer be rejected.

Dependent Variable: RDINTENS
Method: Least Squares
Date: 01/06/26 Time: 15:45
Sample: 1 32
Included observations: 32

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.938958	0.458789	4.226255	0.0002
SALES	0.000160	6.46E-05	2.472019	0.0200
PROFMARG	0.070066	0.034326	2.041200	0.0511
DUM_1	6.466894	1.411823	4.580526	0.0001
DUM_10	-5.414306	2.772871	-1.952600	0.0613

R-squared	0.535004	Mean dependent var	3.266210
Adjusted R-squared	0.466116	S.D. dependent var	1.873701
S.E. of regression	1.369065	Akaike info criterion	3.608734
Sum squared resid	50.60714	Schwarz criterion	3.837755
Log likelihood	-52.73974	Hannan-Quinn criter.	3.684648
F-statistic	7.766249	Durbin-Watson stat	1.880845
Prob(F-statistic)	0.000266		





A warning about defining outliers by residual size

Firm 1 has a much larger dummy t-statistic (4.56) than firm 10 (−1.82), so by that measure firm 1 looks like "more of an outlier".

But identifying outliers from the residuals of a regression using all observations is generally a bad idea, precisely because OLS adjusts itself to make those residuals small. Including the largest firm flattened the regression line considerably, which kept its own residual modest – and, given how flat the line then was, made firm 1 with its high R&D intensity look extreme.

The influential observation and the large-residual observation need not be the same one.

4.5 Functional form as a remedy

Certain functional forms are simply less sensitive to outlying observations. For most economic variables the logarithmic transformation substantially narrows the range of the data, and it yields forms – such as constant elasticity models – capable of describing a broader range of data.

Re-estimating in logs, with R&D spending rather than intensity as the dependent variable:

$$\log(rd) = -4.378 + 1.084 \log(sales) + 0.022\ profmarg$$

The elasticity of R&D spending with respect to sales is 1.084 – a little above one, which would mean R&D intensity rises with firm size. But a Wald test cannot reject H_0: $\beta_1 = 1$, so there is not enough evidence to conclude that R&D intensity increases with firm size.

The payoff

Jarque-Bera on the level model rejects normality; on the log model it does not ($p = 0.15$).

The logarithmic transformation has done, through a better-specified functional form, what the dummy variables did by brute force – and it has produced a parameter with a direct economic interpretation to test.

Note that the elasticity being close to one is itself the economic finding: R&D spending is close to proportional to sales.

5. Summary of violations

Assumption	Violation may imply	Possible solutions
MLR.1 Linearity	Wrong regressors; nonlinearity; changing parameters	Transform variables (logs, quadratics); nonlinear estimation methods
MLR.2 Random sampling	Endogenous sample selection	Design sampling so selection does not depend on y; sample-selection corrections for existing data
MLR.3 No perfect collinearity	Multicollinearity (strong imperfect correlation)	Transform variables; drop a variable (risks bias); principal components
MLR.4 Zero conditional mean	Functional form misspecification; omitted variables; reverse causality; measurement error in regressors	Transform variables, use quadratics; proxy variables; instrumental variables; difference-in-differences
MLR.5 Homoskedasticity	Heteroskedasticity	Robust standard errors; generalised least squares
MLR.6 Normality	Outliers; wrong functional form	Transform variables (logs); dummy out or eliminate outliers; rely on large-sample normality

The hierarchy worth remembering

MLR.5 and MLR.6 affect inference only – the coefficients remain unbiased and consistent, and robust standard errors or a large sample repair the damage.

MLR.3 violates nothing; it costs precision, and more data would help.

Assumption	Violation may imply	Possible solutions

MLR.2 (endogenous selection) and MLR.4 (endogeneity) are the serious ones: they bias the coefficients themselves, and no sample size rescues them.

That is why Lecture 7 is devoted entirely to MLR.4.

Key terms

Term	Meaning
Exogenous sample selection	Selection based on the explanatory variables. OLS remains unbiased and consistent.
Endogenous sample selection	Selection based on the dependent variable. Always biases OLS, because selecting on y means selecting on u.
Multicollinearity	High but not perfect correlation among regressors. Violates no assumption; inflates variances.
Variance inflation factor	$VIF_j = 1/(1 - R_j^2)$. The factor by which $Var(\hat{\beta}_j)$ is raised by correlation with other regressors.
Principal components analysis	Replacing correlated regressors with a few uncorrelated linear combinations; reduces collinearity at the cost of interpretability.
Heteroskedasticity	Var(u | x) depends on x. Does not bias OLS, but invalidates the usual standard errors and tests.
Breusch-Pagan-Godfrey test	LM test regressing squared residuals on the original regressors; $LM = n \cdot R^2 \sim \chi^2_k$.
White test	More general LM test adding squares and cross products of the regressors to the auxiliary equation.
Robust standard errors	Consistent standard errors valid with or without heteroskedasticity. Change only the standard errors, never the coefficients.
Jarque-Bera test	Tests normality of the residuals using their skewness and kurtosis.
Asymptotic normality	OLS estimators are approximately normal in large samples even when MLR.6 fails.
Influential observation	An observation whose removal changes the key OLS estimates by a practically large amount.

Review questions

Use these to check your understanding before the seminar.

1. A survey of savings behaviour includes only people over 35. Is OLS biased? Now suppose the survey included only people who saved more than €5,000 last year. Answer again, and explain what changed.
2. Show, using $y = \beta x + u$, why keeping only observations with high y induces $Cov(x, u) \neq 0$.
3. A colleague says "multicollinearity violates MLR.3, so OLS is biased." Correct both halves of this statement.

4. Your model reports VIF = 12 on one regressor. What does this tell you quantitatively about R_j^2, and what – if anything – should you do?
5. Explain why dropping a collinear variable might make things worse rather than better.
6. Explain the etymology of "heteroskedasticity" and give an economic example where you would expect the error variance to rise with a regressor.
7. List what heteroskedasticity does and does not affect. Why does a larger sample not solve the problem?
8. Set out the five steps of the Breusch-Pagan-Godfrey test. What exactly is the null hypothesis, and what distribution does the statistic follow?
9. How does the White test differ from Breusch-Pagan-Godfrey, and what is the practical cost of its greater generality?
10. After switching to robust standard errors your coefficients are unchanged but two t-statistics fall. Explain why the coefficients did not move.
11. If robust standard errors are valid more often, why are the usual ones still reported? Under what circumstances would you prefer them?
12. Jarque-Bera rejects normality with p = 0.00 in a sample of 526. How worried should you be? Would your answer change with n = 32?
13. Explain why OLS is especially sensitive to outliers, referring to the objective function it minimises.
14. In the R&D example, adding a dummy for the largest firm more than triples the coefficient on sales. Explain what the dummy is doing and why its coefficient of −6.57 is informative.
15. Firm 1 has a dummy t-statistic of 4.56 and firm 10 of −1.82. Does this mean firm 1 is the more influential observation? Explain the distinction.
16. The log model gives an elasticity of 1.084 that is not significantly different from one. State precisely what this implies about the relationship between R&D intensity and firm size.
17. Rank MLR.2 (endogenous selection), MLR.3, MLR.5 and MLR.6 by how serious their violation is, and justify your ranking in terms of bias versus precision versus inference.

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 8 and 9 cover the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapters 5, 6 and 8.